

3.5 Derivatives of Trigonometric Functions

Many phenomena of nature are approximately periodic (electromagnetic fields, heart rhythms, tides, weather). The derivatives of sines and cosines play a key role in describing periodic changes. This section shows how to differentiate the six basic trigonometric functions.

① Derivative of the Sine Function

With the help of the angle sum formula for the sine function,

$$\sin(x + h) = \sin x \cos h + \cos x \sin h,$$

we get the following formula.

■ The derivative of the sine function is the cosine function:

$$\frac{d}{dx}(\sin x) = \cos x.$$

Example 1 Find derivatives of the sine function involving differences, products, and quotients.

(a) $y = x^2 - \sin x$.

(b) $y = x^2 \sin x$.

(c) $y = \frac{\sin x}{x}$.

② Derivative of the Cosine Function

With the help of the angle sum formula for the cosine function,

$$\cos(x + h) = \cos x \cos h - \sin x \sin h,$$

we have the following formula.

■ The derivative of the cosine function is the negative of the sine function:

$$\frac{d}{dx}(\cos x) = -\sin x.$$

Example 2 Find derivatives of the cosine function in combinations with other functions.

(a) $y = 5x + \cos x.$

(b) $y = \sin x \cos x.$

(c) $y = \frac{\cos x}{1 - \sin x}.$

③ Derivatives of the Other Basic Trigonometric Functions

Because $\sin x$ and $\cos x$ are differentiable functions of x , the related functions

$$\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \text{and} \quad \csc x = \frac{1}{\sin x}$$

are differentiable at every value of x at which they are defined. Their derivatives are calculated from the Quotient Rule.

Example 3 Find derivatives of $y = \tan x$.

Example 4 Find derivatives of $y = \cot x$.

Example 5 Find derivatives of $y = \sec x$.

Example 6 Find derivatives of $y = \csc x$.

■ The derivatives of the other trigonometric functions:

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$